

PRAKRITI CHAND

Graduate Researcher | Plant–Microbe Interactions | Plant Stress Physiology| Plant Molecular Biology| Agricultural Sciences

Location: Columbia, MO, USA

Contact: pc622@missouri.edu

PROFESSIONAL SUMMARY

Graduate researcher with international research experience spanning, plant–microbe interactions, plant molecular biology and agricultural science across Nepal and the United States. Current work focuses on arbuscular mycorrhizal (AM) fungi-mediated nutrient transport and its role in biological nitrogen fixation and carbon allocation in legume, *Medicago truncatula*. Skilled in molecular and laboratory techniques, field research, and data analysis, with a track record of research assistantships, publications, conference presentations, and research project coordination.

CORE COMPETENCIES

Category	Technical Skills
Molecular Biology	Tissue culture, gel electrophoresis, spectrophotometry, RNA/DNA extraction, qPCR, host–microbe interaction monitoring, microorganism identification
Plant Science & Agriculture	Plant–microbe interactions, arbuscular mycorrhizal fungi, plant pathology, horticulture, disease and stress severity screening and scoring, sustainable agriculture, mycorrhizae, rhizobium
Data Analysis & Software	Statistical analysis, data visualization, experimental design; MS Word, MS Excel, GraphPad Prism, R
Field & Research Skills	Field data collection, sample processing, phenotyping, farm and production operations management, project coordination
Analytical Tools	Stereo microscopy, Gas chromatography, 13C Labeling

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Missouri, Columbia, MO | Aug 2024 – Present

- **Nutrient Transport:** Determining the role of AM fungi-mediated nutrient transport in biological nitrogen fixation in *Medicago truncatula*
- **Carbon Allocation:** Analyzing carbon flux in *Medicago truncatula* in response to nutrients supplied through arbuscular mycorrhizal (AM) fungi
- **Plant–Microbe Interactions:** Studying beneficial plant–microbe interactions in the context of plant stress physiology

Research Visiting Scholar

University of Missouri, Columbia, MO | June 2023 – May 2024

- **Plant–Microbial Interactions:** Conducted research on plant–microbial interactions in support of ongoing lab studies
- **Lab Management:** Assisted with general laboratory management practices
- **Greenhouse Operations:** Managed the greenhouse at the East Campus Plant Growth Facility

Research Assistant Intern

National Gene Bank, NARC, Lalitpur, Nepal | Dec 2020 – Apr 2021

- **Disease Screening:** Conducted research on "Screening of 35 Native Genotypes of Mustard Species Against Different Diseases," including daily screening, scoring, and record maintenance across 20 plots (16 plants per plot) to identify resistant, high-yielding varieties
- **Pathogen Identification:** Performed laboratory identification of disease infestations under microscopic examination, including microorganism identification, under the guidance of senior scientists

Survey Assistant Intern

National Gene Bank, NARC, Lalitpur, Nepal | Dec 2020 – Apr 2021

- **Biodiversity Survey:** Conducted research on "Diversity and Use of Neglected, Underutilized, and Wild Horticultural Species" at Godawari, Nepal
- **Species Identification:** Identified 95 species in the study area and 23 species in local markets as potential future sustainable food-security species through a field survey of 100 respondents

Research Associate

Agricultural Technology Centre, Jwagal, Lalitpur, Nepal | Aug 2019 – Sept 2019

- **Soil & Sample Analysis:** Assisted with laboratory analysis of plant, soil, and compost samples; conducted routine soil testing (pH, phosphorus, potassium, organic matter, and texture) on samples submitted from across the country

PROFESSIONAL EXPERIENCE

Farm Production Assistant

Jaya Kalyan Bagbani Kendra, Salyan, Nepal | Jan 2022 – Apr 2023

- **Production Operations:** Supported orchard and production operations across the full crop cycle — transplant and soil preparation, grafting, disease and pest management, harvesting, and postharvest handling — for citrus, mandarin, guava, persimmon, lemon, and walnut crops
- **Farm Planning:** Helped develop seasonal and weekly farm plans, including tillage and fertilizing schedules, and guided farm workers in adopting new agricultural practices

Teacher, Science and English

Shree Sharada Basic School, Salyan, Nepal | May 2021 – Dec 2021

- **Instruction:** Taught Science and English full-time at a government school
- **Student Support:** Monitored and supported students' academic and personal development

EDUCATION

- **M.S. in Plant Sciences & Technology** | University of Missouri, Columbia, MO — Expected Dec 2026, GPA: 3.9
Thesis: Investigating how arbuscular mycorrhizal (AM) fungi mediate nutrient transport to support carbon allocation and biological nitrogen fixation in Medicago truncatula
- **B.S. (Hons) in Agriculture, Horticulture** | Himalayan College of Agric. Science & Technology (Purbanchal University), Kirtipur, Kathmandu, Nepal — GPA: 3.68

AWARDS & ACHIEVEMENTS

- **Daniel Millikan Travel Award (May 2026):** \$1,500 in conference travel support
- **DPST Travel Award (July 2026):** \$600 in conference travel support
- **Plant Science Saturday Award (July 2026):** \$50

KEY PUBLICATIONS

- **Nepalese Journal of Agricultural Sciences (Vol 21):** Chand P. Diversity and Use of Neglected and Underutilized Species at Godawari Municipality, Nepal.

CONFERENCE PRESENTATIONS

- **Plant Biology 2026, ASPB & Canadian Society of Plant Biologists (Ottawa, Canada, July 2026):** Poster presentation.
- **ASPB Midwest Section Annual Meeting (April 2026):** Poster presentation.
- **Agricultural Biodiversity Conference, WASET Vol 15 No. 09 (Rome, Italy, Sept 2022):** Selected for oral presentation.

CERTIFICATIONS AND TRAINING

- Sustainable Agricultural Land Management — University of Florida; Communication Strategies for a Virtual Age — University of Toronto
- Organic Inspection and Organic Certification; Vermicomposting — IAAS Nepal
- Composting, Vermicomposting and Rooftop Farming Management Training — Nature Conservation and Sustainable Development Society
- Plant Clinic, IPM Programs for Small Farmers, Poultry Production, Mushroom Cultivation, and Plant Propagation and Nursery Management — Himalayan College of Agricultural Sciences and Technology

LEADERSHIP AND INVOLVEMENT

- Symposium Event Coordinator (April 2025) and Volunteer (March 2024), MU Plant Research Symposium, University of Missouri
- Active Member, IAAS LC Hicast, Amnesty International Nepal Youth Network (Secretary/Organizer), Land Our Future, Youth Advocacy Network, Help Nepal Network, National Youth for Environment Coalition, and Interact Club
- Program Host and Coordinator, National Mass Media in Communications, Nepal — hosted television programs on travel, entertainment, and social outreach across MTH Channel, Kollywood TV, and Chalachitra Khabar (2017–2019)